

Statistics and Probability

26FS

Programme	BSc in Business Information Technology
Degree	Bachelor
ECTS	5
Module type	mandatory module
Module coordinator	Dr. Souvik Datta
Leading principle / Short description	Data analysis and basic statistical concepts have become indispensable in everyday professional life. This course is designed to enable students to correctly classify and evaluate statistical variables in relation to their level of measurement. With acquired knowledge of probability calculations, distributions and statistical tests, evaluations and analyses can be correctly commented on and, if necessary, critically questioned.
Module content	- o Probability calculus - o Random experiment, definitions of probability, events, exclusive and independent events, laws of addition and multiplication, conditional probability, Bayes theorem, combinatorial analysis, random sampling - o Descriptive statistics - o Measurement levels, absolute and relative frequencies, frequency tables, graphics, empirical distribution function, location parameters (mode, median, mean), dispersion parameters (range, standard deviation, variance, coefficient of variation, interquartile range), quantiles. - o Probability distributions - o Random variables and distributions (binomial and normal distribution) - o Linear correlation and linear regression - o Correlation coefficient, spurious correlation, linear regression - o Analytical statistics - o Point estimation and standard error, confidence interval, checking distributional form, significance tests (t- test for independent samples), contingency tables (chi- squared test, measures of association). - o Statistical analysis - o Statistical analysis with SPSS, R or Excel
Competencies to be achieved	Knowledge and Understanding: <i>Students ...</i> - o know how to classify statistical variables and the implication of different measurement levels on data evaluation techniques - o know basic visualization techniques and descriptive techniques - o know the differences between univariate and bivariate analysis - o know different statistical distributions of a quantitative variable (Normal, t, Poisson, Binomial and others) - o understand the difference between parametric and nonparametric methods - o learn different statistical tests and understand their limitations Applying Knowledge and Understanding: <i>Students ...</i> - o can provide summary statistics on a statistical variable of any measurement level - o can describe statistical variables using appropriate visualisation techniques for both univariate and bivariate techniques - o can interpret the results of statistical tests and comment on the limitations of these results Making Judgements: <i>Students ...</i> - o can use statistical techniques to derive judgements on significance of differences in variables - o can read quantitative reports and comment on them Communications Skills: <i>Students ...</i> - o can transfer results of a statistical analysis to a nonexpert audience Learning Skills: <i>Students ...</i> - o can transfer theory to practice by using appropriate statistical software for data analysis - o develop initiative by oneself to invest time in knowledge acquisition related to data analytics outside the course
Prerequisites	Elementary Algebra (equations with one variable)
Teaching and learning methods	Contact Hours: Lecture, Discussion, Assignment, Learning checks Guided Self-Study: Individual work, Group work

Literature

Required Reading:

None

Recommended Reading: Sharpe, N. R., De Veaux, R. D., Velleman, P. (2021)
Business Statistics, Global Edition (4ed), Pearson International.**Remarks**

Links to other modules: Business Maths 1

Grading

Grade 1-6 (half grades)

Assessment**End-of-semester examination 100%**

Oral / Written

Written

Duration (min)

90

Timeframe

Exam period (end of Semester)

Grading Scale

Points

Module details**Statistics and Probability (BITFT 2a) - Tue - Basel**

Time	1:15 PM - 5:00 PM
Language	Englisch
Max. participants	30
Periodicity	Weekly
Lecturers	Prof. Dr. Tobias Schoch
Number	2-26FS.W-B-BITFT-040113S1.SN/FT24

